Workshop construction plan

Synthetic workshop 1500

Transverse laminated wood

Overall assembled length: 1500.0 mm

Internal mouthpiece diameter: 30.6 mm

Internal bell diameter: 83.0 mm

Target note / frequency: C#2 / 69.3 Hz

Predicted passive resonances: 78.2 / 189.1 / 285.0 / 398.2 / 512.3 Hz

Minimum remaining wall: 8.0 mm; Radial overcut reserve: 1.0 mm

Quantity: 30; Square stock (mm)

Stack numbered layers from mouthpiece to bell. Centre all openings and align witness marks. Constant openings approximate the continuous ideal profile; listed lengths exclude glue thickness.

Dimensions rounded to 0.1 mm for marking; not a guaranteed manufacturing tolerance.

Cut list - numbered from mouthpiece

Order	Start-end (mm)	Length (mm)	Opening (mm)	Square stock (mm)
1	0.0-50.0	50.0	30.6	48.7
2	50.0-100.0	50.0	31.3	49.3
3	100.0-150.0	50.0	31.9	50.0
4	150.0-200.0	50.0	32.5	50.6
5	200.0-250.0	50.0	33.2	51.2
6	250.0-300.0	50.0	33.8	51.9
7	300.0-350.0	50.0	34.5	52.5
8	350.0-400.0	50.0	35.1	53.1
9	400.0-450.0	50.0	35.7	53.8
10	450.0-500.0	50.0	36.4	54.4
11	500.0-550.0	50.0	37.0	55.1
12	550.0-600.0	50.0	37.6	55.7
13	600.0-650.0	50.0	38.3	56.3
14	650.0-700.0	50.0	38.9	57.0
15	700.0-750.0	50.0	39.6	57.6
16	750.0-800.0	50.0	40.2	58.2
17	800.0-850.0	50.0	40.8	58.9
18	850.0-900.0	50.0	41.5	59.5
19	900.0-950.0	50.0	42.1	60.2
20	950.0-1000.0	50.0	42.7	60.8
21	1000.0-1050.0	50.0	45.5	63.5
22	1050.0-1100.0	50.0	49.7	67.7
23	1100.0-1150.0	50.0	53.8	71.9
24	1150.0-1200.0	50.0	58.0	76.0
25	1200.0-1250.0	50.0	62.2	80.2
26	1250.0-1300.0	50.0	66.3	84.4
27	1300.0-1350.0	50.0	70.5	88.5
28	1350.0-1400.0	50.0	74.7	92.7
29	1400.0-1450.0	50.0	78.8	96.9
30	1450.0-1500.0	50.0	83.0	101.0

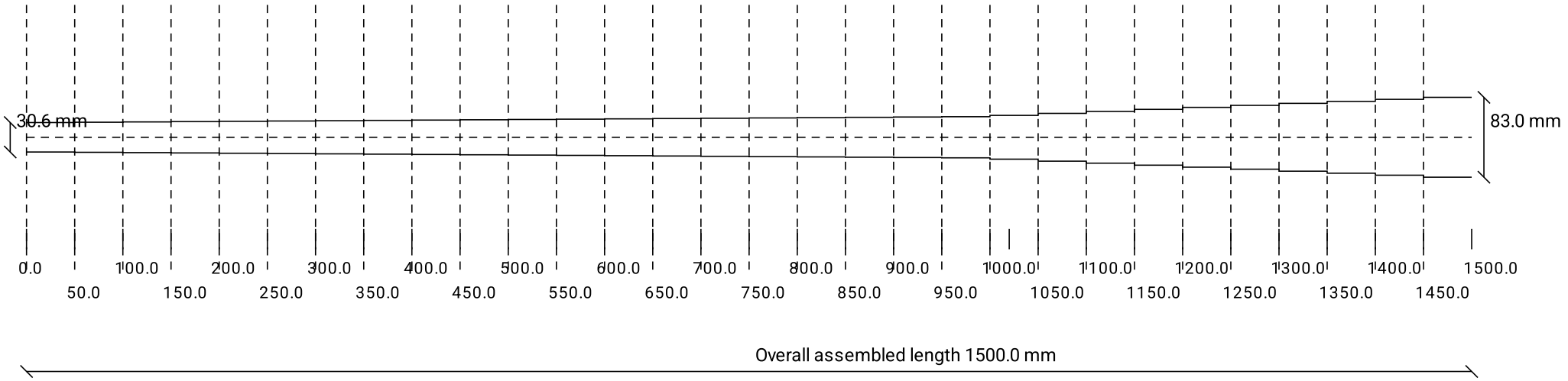
Passive acoustic prediction, not a guaranteed played note. Check material, joints and wall integrity.

Dimensioned construction drawing

Synthetic workshop 1500 - Transverse laminated wood

Longitudinal scale 1:6.0

Internal longitudinal profile



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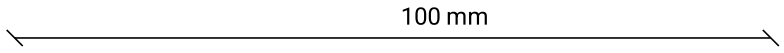
Dimensions rounded to 0.1 mm for marking; not a guaranteed manufacturing tolerance.



100 mm - print at 100% and verify with a ruler